# Dynamic Valuation Timing under Trading Frictions

A Mathematical Reconstruction and Controlled QuantConnect Backtest

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**Abstract**

This report develops a reproducible implementation of dynamic valuation timing with predictable returns and costly portfolio adjustment. A quadratic state representation converts a valuation–macroeconomic interaction model into a finite-dimensional conditional-moment system. The resulting portfolio rule combines the existing exposure with an infinite-horizon aim portfolio obtained from a matrix resolvent. The empirical application allocates between SPY and BIL over January 2020–December 2025 using a rolling 240-month estimation window. The linear valuation specification records a 12.400% compound annual return, a reported Sharpe ratio of 0.740, and an 11.4% maximum drawdown. Its annual return exceeds immediate and gradual 55% equity-allocation controls by 2.834 and 3.528 percentage points, respectively. These are descriptive results for the examined sample, rather than a significance claim. The implementation uses revised CFNAI observations as a macroeconomic proxy and is a methodological reconstruction, not an exact replication of the original PMI or anomaly-factor datasets. Forecast-error and constraint diagnostics delimit the interpretation of the portfolio results.

**Keywords:** valuation timing; dynamic portfolio choice; transaction costs; quadratic state lifting; rolling estimation; controlled backtesting.

## 1 Research question and contribution

Return predictability is useful for a portfolio only when its magnitude, persistence and implementation costs jointly justify changing exposure. The dynamic framework of Gârleanu and Pedersen [2] formalizes partial adjustment toward an aim portfolio. Krogh [1] extends the question to state-dependent predictability; the Huaan Securities research note [3] provides the starting point for this reconstruction.

This project contributes a transparent numerical implementation and an audited case study. It connects a return regression to a state-transition model, derives the adjustment rule, and tests its executed portfolio against controls that distinguish varying exposure from a fixed allocation and from gradual entry. The propositions below give self-contained derivations of the model implemented in the code; they are not presented as new theoretical discoveries.

The empirical object is an aggregate equity timing signal, rather than a new cross-sectional stock-selection factor. The primary result concerns the linear valuation specification. Auxiliary interaction models are retained to identify the roles of nonlinear predictors, sign filters and exposure limits.

# 2 Signal construction and information timing

## 2.1 Valuation and realized returns

Let $P_d > 0$ be the raw SPY closing price on trading day $d$, and let $D_d$ be its cash dividend on the ex-dividend date. The historical return series used for estimation compounds daily dividend-inclusive gross returns:

$$G_d = \frac{P_d + D_d}{P_{d-1}}, \qquad R_m = \prod_{d \in m} G_d - 1. \tag{1}$$

Thus cash dividends are not added again to an already dividend-adjusted price series. Denote by $P_m$ the last available raw close in month $m$, and by $\mathcal{D}_m$ the sum of dividends in the last twelve complete calendar months. The valuation predictor is

$$x_m = \log\left(\frac{\mathcal{D}_m}{P_m}\right). \tag{2}$$

This is a trailing dividend-price signal, not a direct estimate of fundamental intrinsic value.

Let $s_m$ be the CFNAI observation selected by the release-tape information set at month-end. The training target is the next calendar month's equity excess return, $r^e_{m+1} = R_{m+1} - r^f_{m+1}$. A previously available TB3MS quote supplies a monthly financing-rate approximation. TB3MS is quoted on a discount basis; the implemented compounding conversion is a proxy rather than an exact Treasury-bill holding-period return. Actual BIL profit and loss is included separately in the trading account.

## 2.2 Rolling estimation and four return equations

Within each completed training window, standardize valuation and macroeconomic states using only that window's mean and sample standard deviation. Write the standardized state as $f_m = (\widetilde{x}_m, \widetilde{s}_m)^\mathsf{T}$. The four conditional-mean specifications are

$$\begin{aligned}
\text{Mean:} \quad & \mu_m = b_0, \\
\text{Linear:} \quad & \mu_m = b_0 + b_x \widetilde{x}_m, \\
\text{Bivariate:} \quad & \mu_m = b_0 + b_x \widetilde{x}_m + b_s \widetilde{s}_m, \\
\text{Interaction:} \quad & \mu_m = b_0 + b_x \widetilde{x}_m + b_s \widetilde{s}_m + b_{xs} \widetilde{x}_m \widetilde{s}_m.
\end{aligned} \tag{3}$$

Each regression uses 240 completed $(f_m, f_{m+1}, r^e_{m+1})$ observations. The label "linear" refers to the return equation. Every specification retains the same two-state VAR; the macroeconomic state can therefore influence the linear strategy's long-horizon aim indirectly through future valuation dynamics.

In the interaction equation,

$$\frac{\partial \mu_m}{\partial \widetilde{x}_m} = b_x + b_{xs} \widetilde{s}_m. \tag{4}$$

A negative interaction coefficient makes the valuation sensitivity stronger in lower macroeconomic states. An optional economic-sign filter requires $b_x > 0$ and, for the interaction specification, $b_{xs} < 0$. A failed sign check replaces the complete return forecast with the historical mean.

## 2.3 An explicit information-set convention

At a first-trading-day decision, the algorithm forms signals from the previous complete calendar month and refreshes historical prices only through completed trading days. Its release tape

selects eligible observations by `available_at`, with a separate row permitted for every historical revision. The supplied snapshots, however, contain revised historical CFNAI and TB3MS values downloaded on 28 October 2026. Their availability is *assumed* to be the first day of observation month plus two months. This convention prevents using a later observation month directly, but does not remove revision bias.

# 3 Dynamic optimization and the adjustment coefficient

## 3.1 The stylized portfolio problem

For the mathematical derivation, let $w \in \mathbb{R}^n$ denote exposures per unit of risk capital, $v$ the inherited exposure, $\mu(f)$ the expected next-period excess-return vector, and $\Sigma \succ 0$ a covariance matrix held fixed within the current estimation window. Let $\gamma > 0$ be risk aversion, $\lambda \geq 0$ the adjustment penalty, and $0 < \beta < 1$ the one-period discount factor. Consider the Bellman equation

$$V(v, f) = \max_{w} \Big\{ -\frac{\lambda}{2}(w - v)^{\mathsf{T}}\Sigma(w - v) + \beta\Big[w^{\mathsf{T}}\mu(f) - \frac{\gamma}{2}w^{\mathsf{T}}\Sigma w + \mathbb{E}[V(w, f') \mid f]\Big]\Big\}. \tag{5}$$

Adjustment occurs now; return and risk contributions enter at the next date. This discounting convention matters for the Riccati coefficient. Assume stable state dynamics and sufficient moments to make the discounted value finite. The exposures are unconstrained in this derivation. Translation into fractions of current portfolio equity and long-only limits is an implementation step.

**Proposition 1** (Partial adjustment)**.** *For $\lambda > 0$, define*

$$B = \gamma + \frac{1 - \beta}{\beta}\lambda, \qquad a = \frac{\sqrt{B^2 + 4\gamma\lambda} - B}{2}, \qquad \kappa = \frac{a}{\lambda}. \tag{6}$$

*Then $0 < a < \lambda$ and $0 < \kappa < 1$. The optimal unconstrained exposure takes the form*

$$w^*(v, f) = (1 - \kappa)v + \kappa\ \mathrm{aim}(f). \tag{7}$$

*Proof.* Use the value-function form $V(v, f) = -\frac{a}{2}v^{\mathsf{T}}\Sigma v + v^{\mathsf{T}}b(f) + c(f)$. Let $h(f) = \mu(f) + \mathbb{E}[b(f') \mid f]$ and $d = \lambda + \beta(\gamma + a)$. The first-order condition in equation (5) is

$$d\Sigma w = \lambda\Sigma v + \beta h(f). \tag{8}$$

Matching the quadratic coefficient in the optimized value function gives $a = \lambda - \lambda^2/d$. Eliminating $d$ yields

$$\beta a^2 + [\beta\gamma + (1 - \beta)\lambda]a - \beta\gamma\lambda = 0, \tag{9}$$

whose positive solution is equation (6). The polynomial is negative at $a = 0$ and positive at $a = \lambda$, so its unique positive root lies between them. Since $\lambda/d = 1 - a/\lambda$, the inherited-exposure coefficient in equation (8) is $1 - \kappa$. Defining the remaining term as $\kappa\,\mathrm{aim}(f)$ establishes equation (7). □

For numerical stability, the implementation evaluates the equivalent root

$$a = \frac{2\gamma\lambda}{B + \sqrt{B^2 + 4\gamma\lambda}}, \tag{10}$$

which avoids subtracting nearly equal positive numbers when $\lambda$ is large.

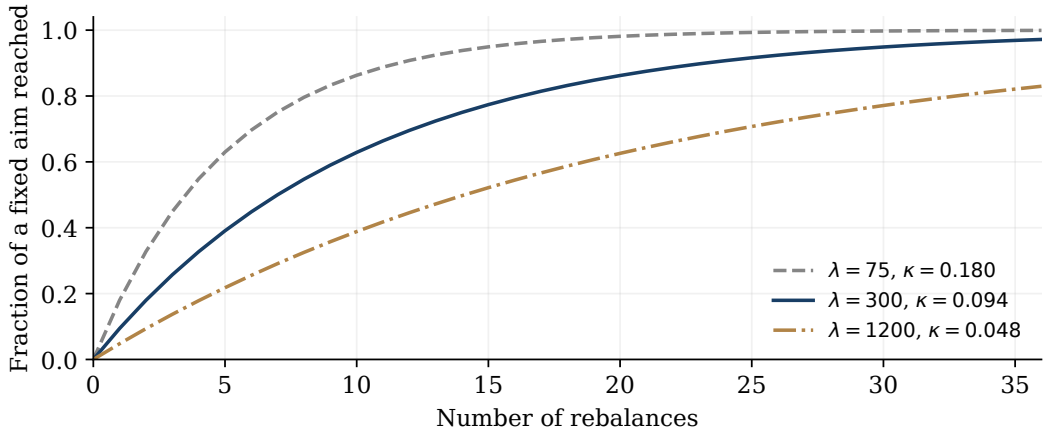



**Figure 1:** Adjustment toward a fixed aim, with no holding drift. Curves show $1-(1-\kappa)^m$ for three illustrative penalties at $\gamma = 3$ and a 2% annual discount rate. This is a theoretical sensitivity illustration, not an additional backtest or a search used to choose the empirical penalty.

### 3.2 Persistence-adjusted aim portfolio

Define the conditional-expectation operator $(\mathcal{P}g)(f) = \mathbb{E}[g(f') \mid f]$ and

$$q = \beta(1-\kappa) = \frac{a}{\gamma + a} \in (0,1). \tag{11}$$

The equality follows by rearranging equation (9).

**Proposition 2** (Infinite-horizon aim)**.** *If the conditional-mean sequence is summable under weights $q^j$, then*

$$\text{aim}(f) = \frac{1-q}{\gamma}\Sigma^{-1}\sum_{j=0}^{\infty} q^j \mathbb{E}[\mu(f_{m+j}) \mid f_m = f]. \tag{12}$$

*Proof.* Matching the linear term in the optimized value function gives $b = qh$. Consequently $h = \mu + q\mathcal{P}h$, and $h = (I - q\mathcal{P})^{-1}\mu = \sum_{j\geq 0} q^j \mathcal{P}^j \mu$. From equation (8), the aim multiplier is $\beta/(d\kappa) = 1/(\gamma + a) = (1-q)/\gamma$. Substitution proves the result. □

The weights $(1-q)q^j$ sum to one. Thus the aim is an average of present and expected future mean-variance targets, rather than the current one-period target alone. At $\lambda = 0$, the limiting rule is $\kappa = 1$, $q = 0$ and $w = \gamma^{-1}\Sigma^{-1}\mu(f)$.

**Corollary 3** (Constant-mean special case)**.** *If $\mu(f) \equiv \overline{\mu}$, then* $\text{aim} = \gamma^{-1}\Sigma^{-1}\overline{\mu}$*. With a fixed aim and no intervening holding drift, the policy satisfies*

$$w_m - \text{aim} = (1-\kappa)^m (w_0 - \text{aim}). \tag{13}$$

The numerical experiment uses $\beta = (1.02)^{-1/12}$, $\gamma = 3$ and $\lambda = 300$. The adjustment speed is approximately $\kappa = 0.09434$: each decision closes about 9.43% of the gap to the current aim, before the exposure projection. Figure 1 isolates this mathematical implication.

## 4 Quadratic state closure and the matrix resolvent

Assume the standardized state evolves according to

$$f_{m+1} = c + Af_m + \eta_{m+1}, \qquad \mathbb{E}[\eta_{m+1} \mid f_m] = 0, \quad \mathbb{E}[\eta_{m+1}\eta_{m+1}^{\mathsf{T}} \mid f_m] = \Omega. \tag{14}$$

For $f = (x, s)^\mathsf{T}$, define the complete quadratic lift

$$z(f) = (1, x, s, x^2, xs, s^2)^\mathsf{T}, \qquad \mu(f) = Cz(f). \tag{15}$$

In the interaction regression, $C = (b_0, b_x, b_s, 0, b_{xs}, 0)$ for one risky return. Zeros in this current-return row do not justify removing the square features from future-state dynamics.

**Proposition 4** (Moment closure and finite-dimensional aim)**.** *Under equation* (14)*, a* $6 \times 6$ *matrix* $T$ *exists such that*

$$\mathbb{E}[z(f_{m+1}) \mid f_m] = Tz(f_m), \qquad \mathbb{E}[\mu(f_{m+j}) \mid f_m] = CT^j z(f_m). \tag{16}$$

*If* $\rho(qT) < 1$*, then the aim is*

$$\boxed{\operatorname{aim}(f) = (\gamma\Sigma)^{-1}(1-q)C(I-qT)^{-1}z(f).} \tag{17}$$

*Proof.* Write the conditional mean of state component $i$ as $\ell_i(f) = c_i + \sum_k A_{ik} f_k$. Its conditional cross moment is

$$\mathbb{E}[f_{i,m+1} f_{j,m+1} \mid f_m = f] = \ell_i(f)\ell_j(f) + \Omega_{ij}. \tag{18}$$

The right side is a quadratic polynomial in $f$, so each lifted component is a linear combination of the six features in equation (15). Together with the constant and first moments, these coefficients define $T$. Iterated conditional expectation proves equation (16). Substituting it into equation (12) and using the matrix geometric series $\sum_{j\geq 0}(qT)^j = (I - qT)^{-1}$ proves equation (17). □

For example, the $xs$ row includes $x^2$ and $s^2$ terms whenever both components load on both states. In the ordered basis above, that row is

$$\begin{aligned} T_{xs,\cdot} = \big(&c_x c_s + \Omega_{xs},\ c_x A_{s1} + c_s A_{x1},\ c_x A_{s2} + c_s A_{x2},\\ &A_{x1}A_{s1},\ A_{x1}A_{s2} + A_{x2}A_{s1},\ A_{x2}A_{s2}\big). \end{aligned} \tag{19}$$

This makes an interaction forecast analytically tractable without discarding the moments generated by the state transition.

For a stable VAR, the lifted transition has eigenvalues associated with the constant, $A$, and pairwise products of eigenvalues of $A$. Hence $\rho(A) < 1$ implies $\rho(T) = 1$, and $q < 1$ gives $\rho(qT) < 1$. The implementation caps the estimated VAR spectral radius at 0.995, floors covariance eigenvalues at $10^{-8}$, and rejects a resolvent with condition number above $10^{12}$. Linear systems are solved numerically; explicit matrix inversion in equation (17) is mathematical notation.

# 5 From the unconstrained solution to executable trades

## 5.1 Monthly policy and residual allocation

For the traded application, the risk-return equation has one risky leg, SPY. The inherited weight is the actual SPY holdings value divided by current portfolio equity. After computing the unconstrained update, the algorithm uses

$$\widehat{w}_m = (1-\kappa)w_{m^-} + \kappa \operatorname{aim}(f_m), \qquad w_m = \min\{0.99, \max\{0, \widehat{w}_m\}\}, \tag{20}$$

with BIL target $0.995 - w_m$. The target allocation reserves 0.5% cash before platform buffers, fees and whole-share rounding. BIL is a tradable short-term Treasury ETF proxy, not an identical replacement for the financing series in the regression.

**Table 1:** Common execution and estimation settings.

| Item | Setting |
|---|---|
| Trading interval | 1 January 2020–31 December 2025 |
| Initial capital | USD 1,000,000 |
| Instruments | SPY; BIL for the residual target allocation |
| Estimation | Rolling 240 completed monthly observation pairs |
| Decision time | First exchange trading day, 10:00 New York time |
| Execution data | Minute-resolution raw equity data |
| Policy parameters | $\gamma = 3$, $\lambda = 300$, annual discount rate 2% |
| Numerical stabilizers | VAR radius cap 0.995; covariance eigenvalue floor $10^{-8}$ |
| Exposure limits | SPY target from 0 to 99%; target total invested fraction 99.5% |
| Realized costs | Interactive Brokers brokerage model; constant 1 bp slippage |
| Diagnostics | 72 target decisions; 71 common forecast-scoring months |

The closed-form optimality result applies to equation (5) before this projection. Clipping an unconstrained dynamic policy is a feasible engineering adaptation, not the exact solution to the corresponding box-constrained dynamic program. Similarly, the risk-capital formulation does not establish optimality for arbitrary terminal-wealth utility.

### 5.2 Policy penalties and realized trading costs

The model penalty $\lambda$ determines how quickly exposure changes. QuantConnect portfolio equity, however, is computed using simulated fills, brokerage fees and the specified slippage model. The theoretical quadratic penalty is not subtracted from that equity a second time. This distinction avoids counting two different cost measures as if they were the same realized expense. Reported *Total Fees* is the platform's brokerage-fee statistic; slippage is also reflected through execution prices.

The monthly forecast target and the executed investment interval are close but not identical. Forecasts refer to full calendar-month returns, while the portfolio starts the monthly adjustment after the first trading day's open. Return prediction and executed-account performance therefore require separate evaluation.

## 6 Controlled portfolio evidence

### 6.1 The role of allocation controls

The first control sets a 55% SPY target immediately and restores it each month. The second uses the same partial-adjustment coefficient as the timing policy, but moves toward a constant 55% SPY aim:

$$w_m^B = (1 - \kappa) w_{m^-}^B + \kappa(0.55). \tag{21}$$

Both use the same instruments, execution schedule and cost configuration. The second control continues to respond to actual holding drift; it is not an exact deterministic exponential path in realized markets.

The 55% level was selected after observing the linear strategy's 54.01% average monthly target. It is therefore a retrospective exposure control, not an independently preregistered benchmark or an exact volatility-matched portfolio. Its gradual variant has a lower sample-average target because it starts from cash.

**Table 2:** Executed portfolio results, 2020–2025.

| Metric | Linear timing | Fixed 55% | Gradual 55% |
|---|---|---|---|
| Cumulative net return (%) | 101.758 | 73.080 | 66.597 |
| Compound annual return (%) | 12.400 | 9.566 | 8.872 |
| Reported Sharpe ratio | 0.740 | 0.435 | 0.482 |
| Annual volatility (%) | 7.85 | 9.28 | 6.99 |
| Maximum drawdown (%) | 11.4 | 19.2 | 13.9 |
| Reported beta | 0.417 | 0.538 | 0.362 |
| Mean monthly SPY target (%) | 54.01 | 55.00 | 47.99 |
| Final equity (USD million) | 2.0176 | 1.7308 | 1.6660 |
| Brokerage fees (USD) | 271.39 | 156.88 | 164.81 |
| Order count | 138 | 121 | 108 |

Source: the archived user-supplied QuantConnect exports. Volatility and beta use the higher-precision portfolio-statistics fields; other reported platform metrics retain their published values. Averages refer to monthly targets, not daily actual exposures. No Sharpe ratio is recomputed as CAGR divided by volatility; platform risk-free-rate conventions apply [6].

Table 2 supports a descriptive portfolio comparison. Relative to immediate fixed allocation, linear timing raises the compound annual return by 2.834 percentage points and reduces maximum drawdown by 7.8 percentage points. Relative to gradual entry, the respective differences are 3.528 and 2.5 percentage points. The ending-equity advantages are about USD 286,777 and USD 351,606. Thus average equity allocation and gradual entry alone do not reproduce the timing portfolio's observed path.

These comparisons do not establish a risk-matched alpha. Linear timing has lower volatility and beta than the immediate control, but higher volatility and beta than the gradual control. The reported Sharpe improvements are 0.305 and 0.258, respectively, and remain descriptive statistics without an estimated sampling distribution in this project.

## 6.2 Annual behavior and economic interpretation

**Table 3:** Calendar-year portfolio returns (%).

| Year | Linear timing | Fixed 55% | Gradual 55% |
|---|---|---|---|
| 2020 | 19.645 | 10.198 | 8.466 |
| 2021 | 11.780 | 15.479 | 13.389 |
| 2022 | -4.481 | -9.392 | -9.113 |
| 2023 | 20.139 | 16.367 | 15.536 |
| 2024 | 18.811 | 16.148 | 16.158 |
| 2025 | 10.647 | 11.062 | 11.054 |

Computed from the exported year-end equity observations using New York dates. Values are arithmetic year-over-year percentage changes in account equity; they are not annualized monthly averages.

The exposure path helps explain the aggregate results. Linear timing's mean SPY target is approximately 36.1% in 2022 and 74.9% in 2023. Its 2022 loss is consequently smaller than either fixed-allocation control's, while its 2023 return is higher. This pattern is consistent with useful exposure variation in the examined sample. It is not uniformly dominant: the controls outperform in 2021 and 2025. Keeping these years visible distinguishes a state-dependent allocation mechanism from an assertion of universal superiority.

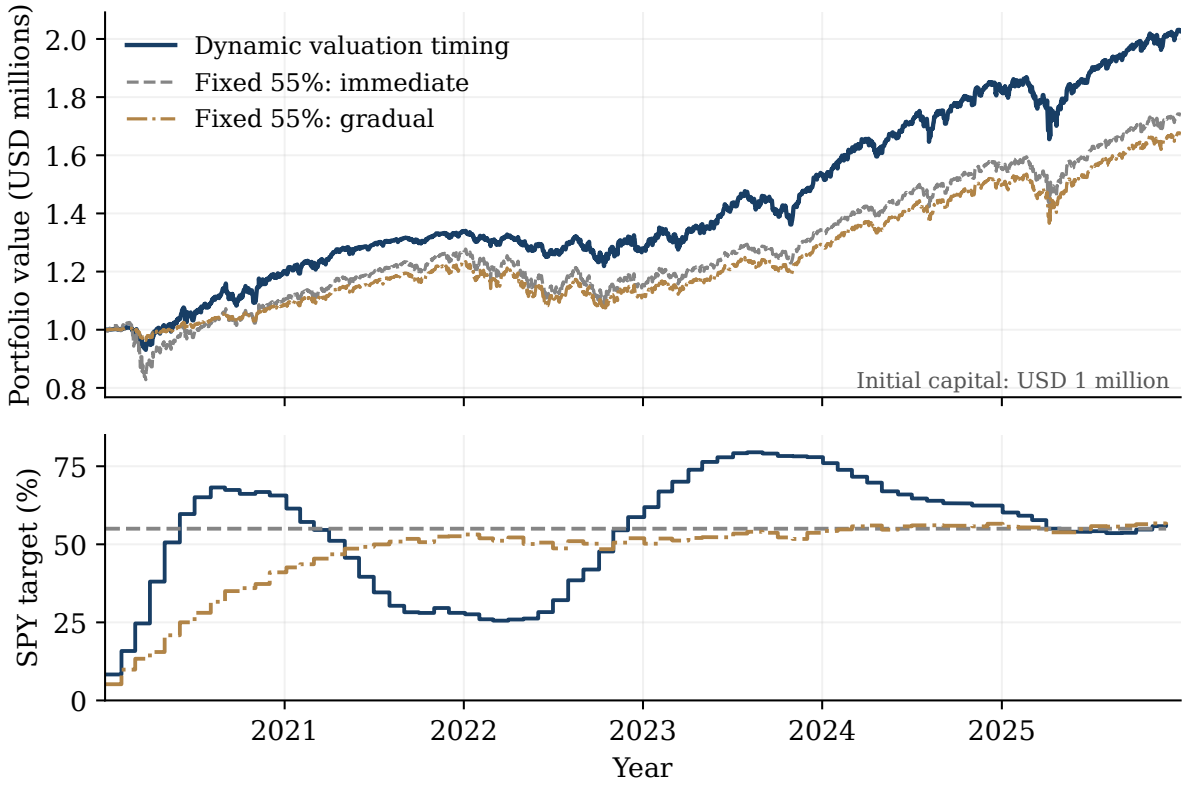



**Figure 2:** Executed account equity and monthly SPY targets. The top panel uses the closing component of the exported equity chart, normalized to the common initial capital. The lower panel shows target weights rather than continuously observed holdings. All three portfolios face the same configured brokerage and slippage models.

## 7 Forecast diagnostics and model-selection discipline

One-step forecast accuracy is evaluated against the contemporaneously estimated historical-mean forecast:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_m (r^e_m - \widehat{r}^e_{m,\text{model}})^2}{\sum_m (r^e_m - \widehat{r}^e_{m,\text{mean}})^2}. \tag{22}$$

The linear specification records $-0.7606\%$ on 71 common scored months. Thus its favorable controlled portfolio result is not evidence that it has beaten the historical mean in squared forecast error. The loss function in equation (22) differs from the risk-sensitive, persistence-adjusted allocation objective; portfolio constraints and the timing of exposure also affect investment outcomes.

Auxiliary diagnostics identify why a nominally nonlinear model need not trade as an unrestricted interaction forecast. With the economic-sign filter on, the interaction model falls back to the mean in 52 of 72 fits and reaches the 99% SPY target cap in 34 decisions. Its mean target is 83.68%. With the filter off, the mean target is 38.43%, no decision reaches the cap, and five targets are zero. The forecast $R^2$ changes from $-0.6183\%$ to $-20.7547\%$. This is evidence that the mean-fallback rule improves forecast-error behavior in this sample; it does not establish the sign restriction as universally valid.

The constrained and diagnostic interaction runs have identical portfolio statistics and target paths. A second download of the final ablation is also a duplicate export. Neither is an additional independent performance sample. The complete model comparison is supplied in table 4.

# 8 Research interpretation and reproducibility

Three aspects are directly supported. First, a complete quadratic moment system makes the dynamic interaction specification computationally explicit. Second, the code implements the distinction between the infinite-horizon aim and partial adjustment toward that aim. Third, the linear specification's observed portfolio path is stronger than both allocation controls on compound return, reported Sharpe and maximum drawdown over the examined interval.

The scope of this evidence is specific. CFNAI substitutes for the original PMI state, the macro history is revised, and the account trades an ETF proxy rather than the paper's original factor portfolios. Actual historical PMI release vintages, original value spreads and the 50-anomaly panel were not obtained. HML/PC1 research interfaces in the code therefore do not constitute completed HML/PC1 empirical replications.

The selected linear baseline and the 55% control reflect information already seen in this six-year evaluation. No untouched holdout, formal strategy-return significance test, or original-paper CER bootstrap was completed. Brokerage assumptions and the policy penalty were also not jointly calibrated to prove realized-cost optimality. The appropriate conclusion is a promising *sample-specific implementation result*, accompanied by explicit mechanism and data checks, rather than a validated forecast of future alpha.

The project archive contains readable Python modules, macro snapshots, configuration exports, original backtest results, diagnostic screenshots, derived tables and checksums. Twenty-six local numerical and data-integrity tests passed. A further comparison of the previous and diagnostic adapters matched targets, forecasts and scoring records over 24 synthetic rebalances. These checks validate implementation properties; the local test harness does not execute LEAN or certify real brokerage fills. Historical cloud outcomes are documented by the user-supplied completed backtest exports.

# 9 Conclusion

The mathematical reconstruction provides a coherent route from a valuation signal to executed portfolio decisions: estimate return and state dynamics, close the system with quadratic moments, solve for a persistence-adjusted aim, and adjust current exposure at an analytically determined speed. The controlled application's principal economic result is a 12.400% compound annual return, a reported Sharpe ratio of 0.740 and an 11.4% maximum drawdown for linear valuation timing. Its improvement over both allocation controls supports retaining that specification as the project's research baseline. The result is reproducible as an archived proxy experiment; its generalization and strict original-data replication remain separate empirical questions.

# A Complete specification comparison and audit notes

The research baseline uses `model=linear`, `economic_constraints=true`, `window=240`, $\gamma = 3$, $\lambda = 300$, `var_cap=0.995` and `max_weight=0.99`. Historical configurations remain in the archive; this report does not alter the underlying trades or regenerate favorable cloud outcomes.

For a single return and two states, the largest resolvent solved is only $6 \times 6$. The same construction extends to $d$ states using $1 + d + d(d+1)/2$ lifted features. The expanding feature dimension is an implementation trade-off, while the covariance and state-transition assumptions continue to determine whether the closed system describes a credible market model.

**Table 4:** Archived specifications and forecast diagnostics.

| Specification | CAGR (%) | Sharpe | MDD (%) | OOS $R^2$ (%) |
|---|---|---|---|---|
| Historical mean | 13.043 | 0.527 | 26.0 | 0 (definition) |
| Linear valuation | 12.400 | 0.740 | 11.4 | -0.7606 |
| Bivariate | 9.601 | 0.558 | 11.6 | -22.3068 |
| Interaction, filter on | 13.676 | 0.577 | 26.0 | -0.6183 |
| Fixed 55%, immediate | 9.566 | 0.435 | 19.2 | n/a |
| Fixed 55%, gradual | 8.872 | 0.482 | 13.9 | n/a |
| Interaction, filter off | 9.027 | 0.543 | 11.5 | -20.7547 |

Seven distinct specifications correspond to eight logical experiment records: the interaction diagnostic rerun repeats the filtered interaction policy. A duplicated download is not counted as another experiment. Initial forecast $R^2$ values are recorded in supplied log screenshots and corroborated by later runtime statistics. Monthly forecast residuals were not separately exported, so no retrospective significance calculation is fabricated from account equity.

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